Zhao Mingze

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Objective

I am actively seeking research assistant, research internship, PhD, and such opportunities to deepen my expertise in AI and their applications in real-world scenarios. With a strong foundation in machine learning and a growing passion for reinforcement learning, I am eager to contribute to cutting-edge research projects while developing my technical and analytical skills. My long-term goal is to pursue a PhD in computer science. Although I am currently a Taught Postgraduate (TPG) student with limited opportunities to engage in research and minimal publications to date, I have never ceased seeking collaborative projects and am deeply committed to continuous learning.

Education

- **The University of Hong Kong** *Sep. 2025 - In Progress*
  Master of Science in Computer Science Hong Kong, China
  - Graduate with distinction.
  - GPA: 3.641/4.0, NLP(A), Quantum Computing(A-), ML & RL(A-), Data Mining(A), AIoT(A).
- **Tongji University** *Sep. 2021 - Jun. 2025*
  Bachelor of Engineering in Software Engineering Shanghai, China
  - GPA: 85/100, Third-class Merit Scholarship, School-level, Dec. 2022.

Research Experiences

- **Unifying Temporal and Structural Credit Assignment in LLM MAS Prompt Optimization [🌐]** *Sep. 2025 - Aug. 2026*
  Research Assistant Shanghai, China
  - Drew the main table and the methodology figure. Drafted Introduction, Related Works, Methodology, Experiments and Conclusion.
  - Complete 90% experimental verification. Deployed vllm using docker running 7-9B LLMs.
- **BIRD-CRITIC NL2SQL Dataset Alignment Annotation [🌐]** *Sep. 2025 - Nov. 2025*
  Research Assistant Hong Kong, China
  - Improve data quality of NL2SQL benchmark BIRD. Rewritten 300+ problematic SQL queries.
  - Checked over 1500 instances systematically, identified and classified 6 common categories of data quality issues.
- **AI-Powered Interview Preparation Platform** *Jan. 2026 - Aug. 2026*
  Postgraduate Project Hong Kong, China
  - Integrating Intelligent Agents, Online Judge, and Personalized Learning.
- **A Large Model for Reasoning and Decision-Making based on Process Reward Models** *Nov. 2024 - May. 2025*
  Undergraduate Thesis Shanghai, China
  - Used LLaMA-Factory to perform SFT and RLHF on Qwen2.5-7B-Instruct.

Internships

- **Cyber Security Company Anxinsec [🌐]** *Jul. 2026 - In Progress*
  AI Dev Intern Beijing, China
  - Responsible for the development of AI agent for the Red Team automated penetration testing platform.
  - Integrated harness engineering methods, improving efficiency by 10% -50%.
- **NEV Company NIO Inc. [🌐]** *Jul. 2024 - Oct. 2024*
  Low-Code Development Intern Shanghai, China
  - Developed a supply chain inventory analysis system, improving computing efficiency by 120%.
  - Developed a backend in Nodejs, a frontend in JavaScript, and a database in MySQL.
  - Conducted forecasts to validate future inventory.
- **Cyber Security Company Qianxin [🌐]** *Jun. 2024 - Jul. 2024*
  Network Security Engineer Shanghai, China
  - Developed data-washing Python scripts, achieving 100% accuracy on gambling dataset.
  - Conducted analysis on hardware, identifying criminal evidence, and presenting findings in documentations.

Skills

- **Fields explored:** base models (Transformers), systems (training & inference), RL, ...
- **Awards:** Chinese Mathematical Contest in Modeling 3rd prize
- **Tech stack:** C/C++, Python, JavaScript, MySQL, docker, PyTorch, sklearn, numpy, matplotlib, gradio, vllm, triton
- **Hobbies:** reading, gaming, writing blogs
- **Languages:** Mandarin (Native), English (Fluent), French & German (Beginner), Cantonese (Beginner)
  - IELTS 7 : Reading 8.5, Writing 6, Listening 6.5, Speaking 6
  - GRE 314 : Verbal 147, Quantitative 166, Writing 3.5

PROJECTS

- **Stanford Spring 2026 CS336: Language Modeling from Scratch** *Dec. 2025 - In Progress*
  Skills: PyTorch, nsys, cuda, triton, RL Repo[] Blog[]
  - Trained a tokenizer using TinyStories and OpenWebText datasets. Trained a 0.017B Transformer LLM.
  - Constructed Transformer with RMSNorm, RoPE, SwiGLU.
  - Explored single GPU nsys, activation checkpointing, operator fusion, and triton.
  - Explored scaling law, data filtering, deduplication, SFT, DPO, and GRPO.
- **MIT 6.S184 Introduction to Flow Matching and Diffusion Models 2026** *Dec. 2025 - In Progress*
  Skills: PyTorch, numpy, matplotlib, differential equations Blog[]
  - An exploration of diffusion large language models.
- **Performed experiments for papers during RA** *Sep. 2025 - Aug. 2026*
  Skills: DSPy, vllm, docker Paper[]
  - To solve the credit allocation problem in multi-agent collaboration with large models, where only the final result is available and it is difficult to locate who/which round went wrong.
  - Designed two mechanisms to respectively identify the poorly performing rounds and agents: 1. Aggregation module recording the intermediate status of each round for time credit. 2. Fixed role prompts for structural credit.
  - Used DSPy to optimize prompt, only modify weak parts, and do not modify all prompts.
  - Deployed vllm using docker running 7-9B LLMs. Wrote a script that enabled optimal GPU usage.
  - Collected results, applied data mining, ran ablation tests.
  - Experiments have shown that this method converges faster and performs better than black box optimization, and can improve the accuracy of multi-agent inference. However, the disadvantage is that it relies on complete interaction trajectories and has high overhead for long sequences.
- **Postgraduate project: AI-Powered Interview Preparation Platform** *Jan. 2026 - In Progress*
  Skills: Java, python, Springboot, LangGraph, harness engineering Blog[]
  - Purpose: 1. Analyze and analyze resumes 2. Generate personalized learning plans 3. Generate daily knowledge points and code questions 4. Simulate interview Q&A.
  - Tech stack: 1. RAG - MinIO + Qdraft + ElasticSearch (RRF) 2. Agent - LLM + prompt + tools + skills + MCP + memory + RAG 3. Loop - ReAct, Plan Execution, DAG.
  - System Design: Technical Document $\subseteq$ Technical Stack $\subseteq$ Position $\subseteq$ Job Category. Client with an OJ.
- **A demo AI agent CLI** *Jun. 2026 - Jul. 2026*
  Skills: python, LangGraph, harness engineering Repo[] Blog[]
  - Architecture: 1. Sequential - Serial Router 2. Fanout - Parallel Read Only 3. Tree - Three Layers of Different Permissions 4. DAG - LLM Task Decomposition + Topology Sorting + Dynamic Appending.
  - LangGraph: 1. Checkpoint - graph snapshot + execution location + timestamp 2. HITL - automatic or approval gating + key decision point 3. complexity gating 4. hierarchical model routing.
  - Highlights: 1. Two level compression + anti-shaking 2. Docker sandbox + downgrade 3. RepoMap 4. SQLite memory system 5. Workplace isolation 6. Permission management + retry.
- **COMP7604 Game final project: A 2D platformer rouge-like game** *Jan. 2026 - May. 2026*
  Skills: C#, Unity, claude code Repo[] Blog[]
  - A trial of pure vibe-coding big project, with 1B tokens consumption.
- **COMP7310 AIoT final project: A step counter based on mobile IMU sensors** *Jan. 2026 - May. 2026*
  Skills: PyTorch, numpy, scipy, mobile sensors Repo[]
  - Implemented offline batch processing, real-time online step detection, configurable signal processing algorithm (LPF, Hampel, Savitzky-Golay, CMA, ACF, FSM), and optional LSTM deep learning model.
- **COMP7607A NLP final project: Musicode** *Sep. 2025 - Nov. 2025*
  Skills: PyQT, Vue3, harness engineering Blog[]
  - Team leader, responsible for the idea. Built up a simple role-based multi-agent system for generating MIDI.
  - Other team members implemented intention recognition, frontend, and OCR.
- **COMP7404 ML & RL mid-term project: Junk mail classification system** *Oct. 2025 - Dec. 2025*
  Skills: Scikit-learn, nltk, JupiterNotebook, machine learning Blog[]
  - Collected junk mail data, preprocessed it, performed feature engineering, trained models, and evaluated models.
  - Models trained: Perceptron, Logistic Regression, KNN, SVM, Decision Tree, Random Forest.
  - Metrics evaluated: precision, recall, f-measure, confusion matrix.
- **COMP7404 ML & RL final project: Implementation of a paper and talking on a conference** *Oct. 2025 - Dec. 2025*
  Skills: PyTorch, Scikit-learn, Transformers, machine learning, talking ability Repo[]
  - Original paper: BiScope: AI-generated Text Detection by Checking Memorization of Preceding Tokens. [Arxiv]
  - Modified LLM call, added local LLM inference. Developed a frontend based on their open-source Python scripts.